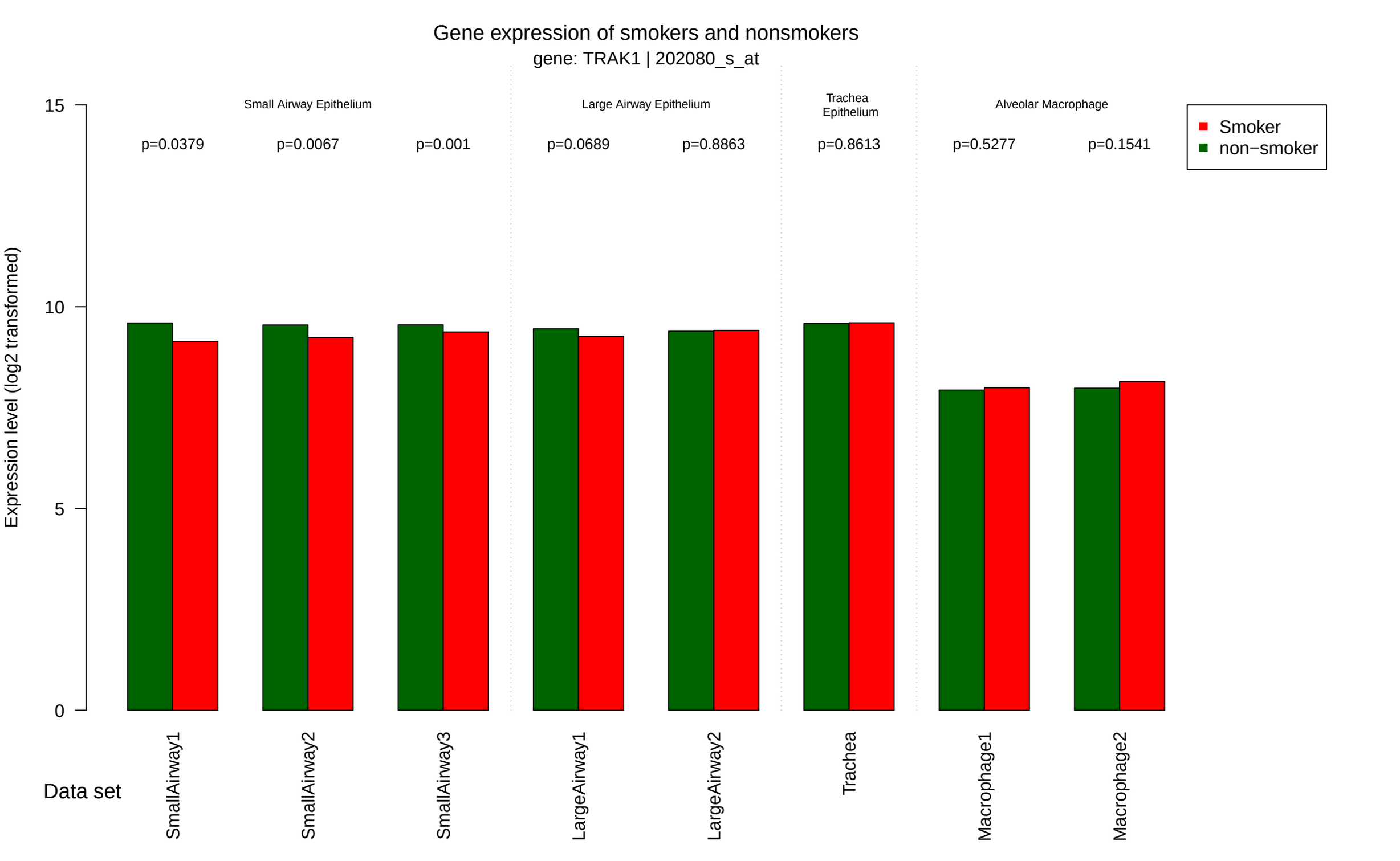
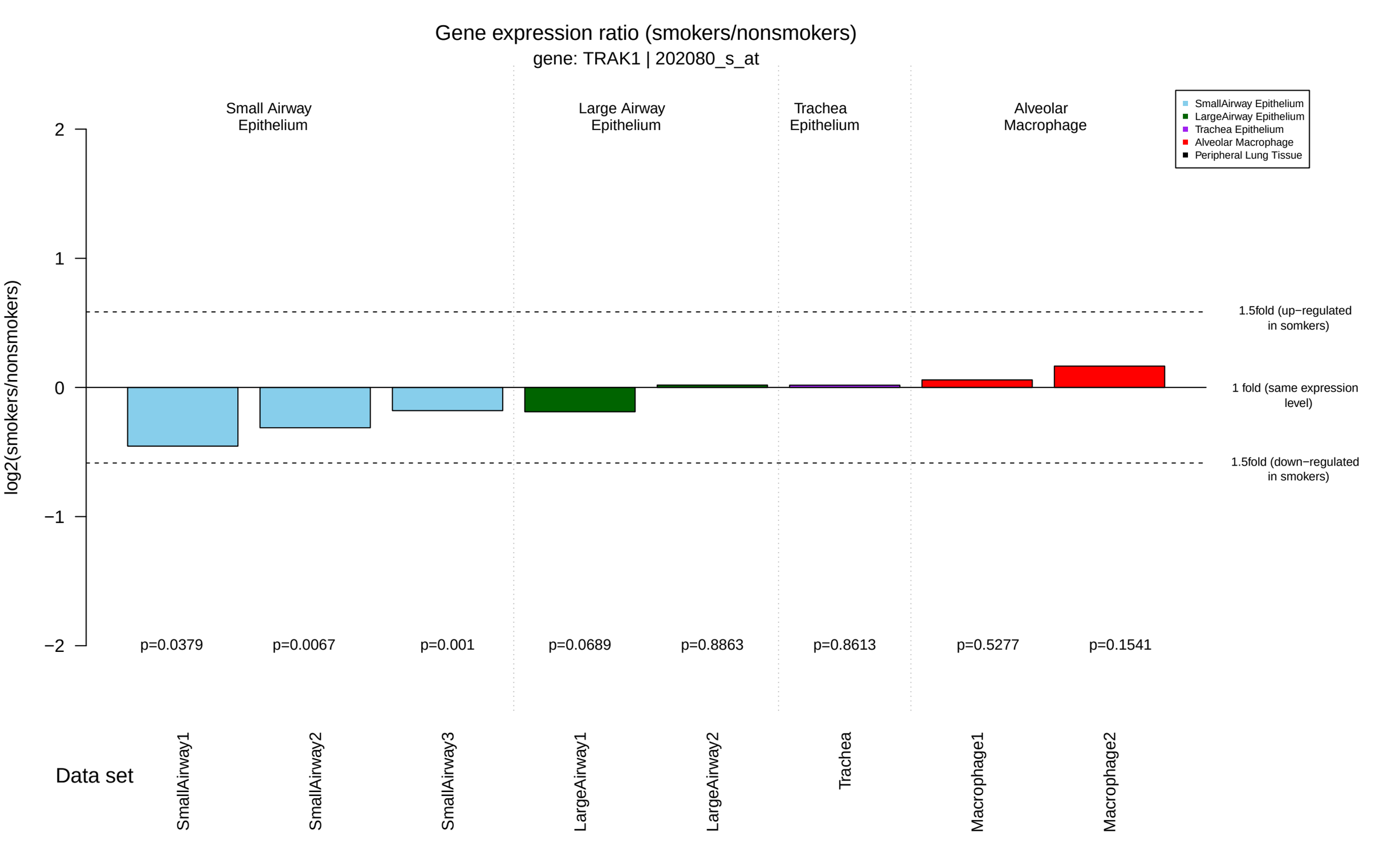
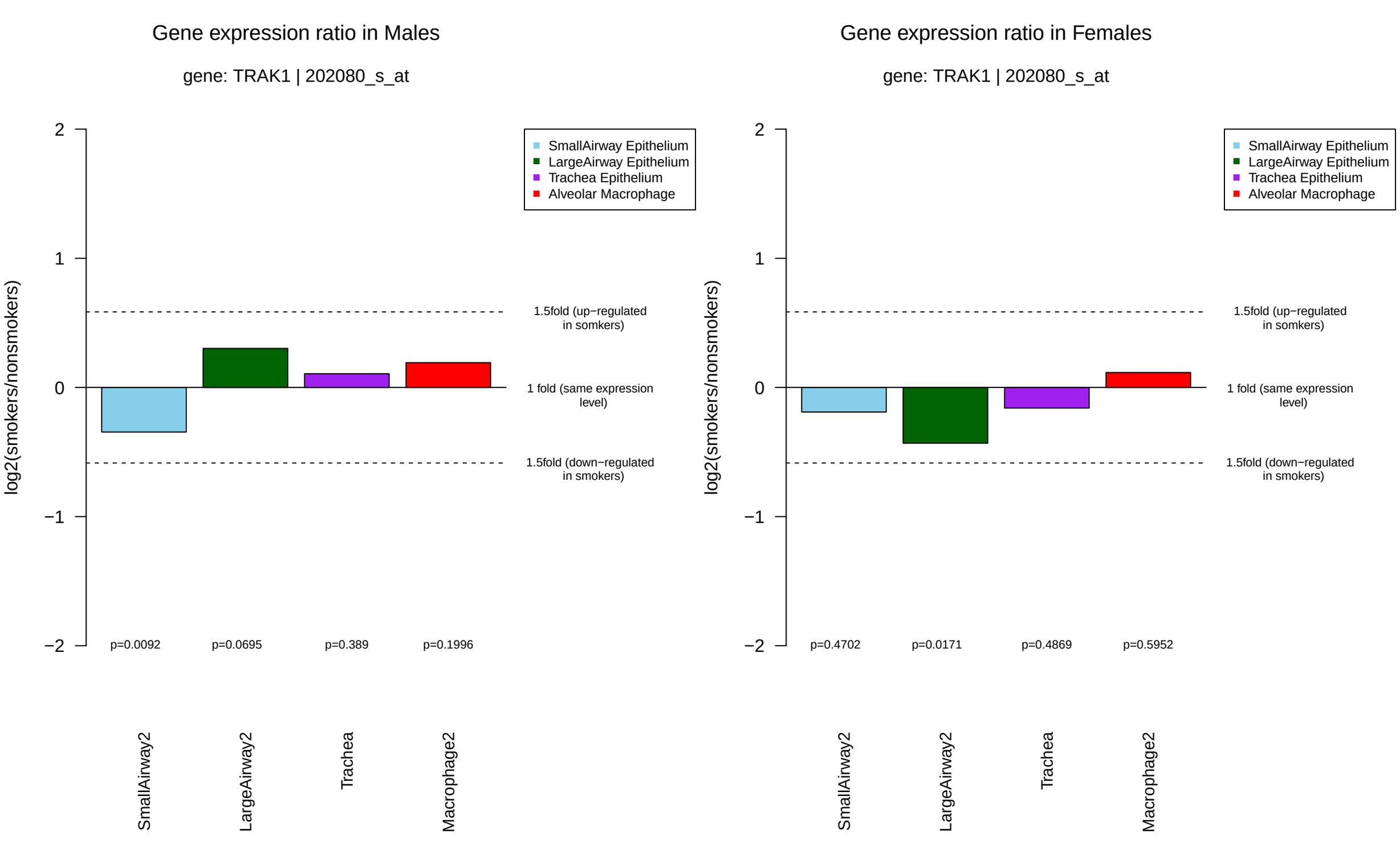
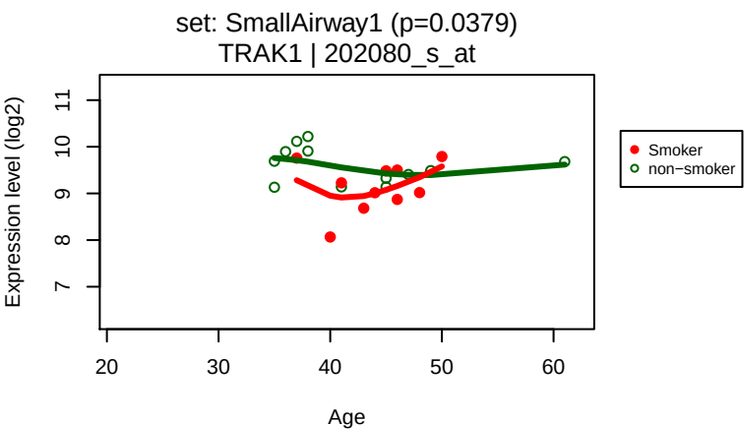
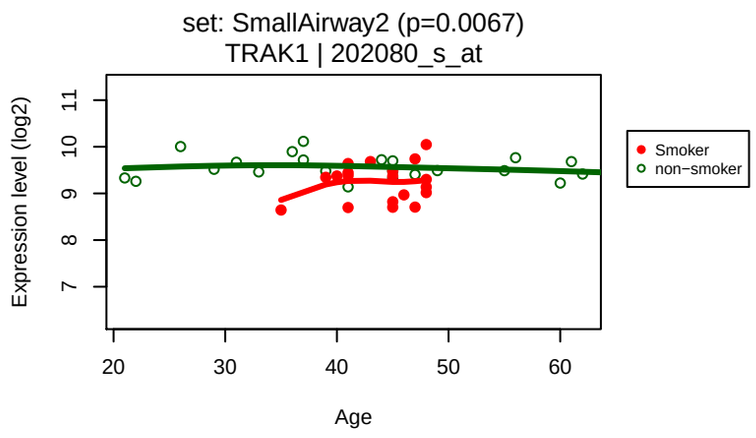
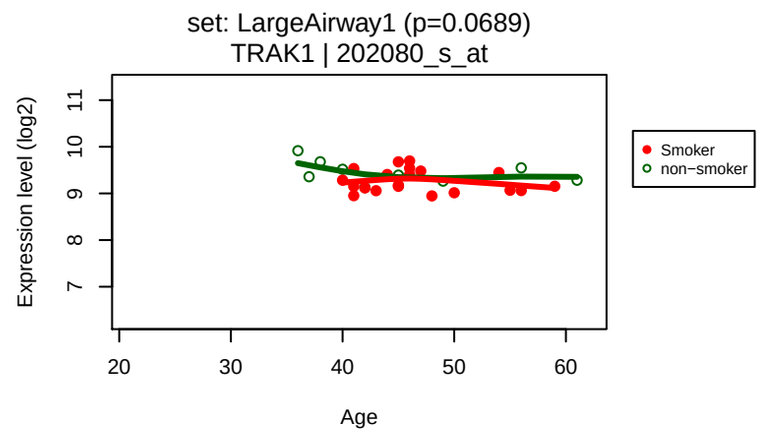

Gene expression of smokers and nonsmokers
gene: TRAK1 | 202080_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0379
p=0.0067
p=0.001
p=0.0689
p=0.8863
p=0.8613
p=0.5277
p=0.1541
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

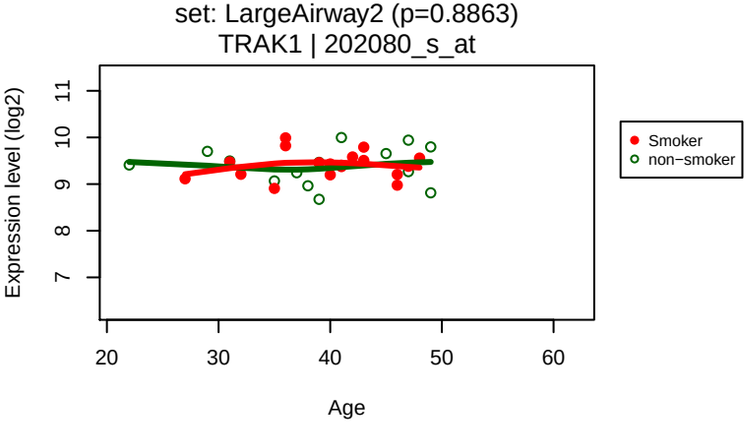

Gene expression ratio (smokers/nonsmokers)
gene: TRAK1 | 202080_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0379
p=0.0067
p=0.001
p=0.0689
p=0.8863
p=0.8613
p=0.5277
p=0.1541
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

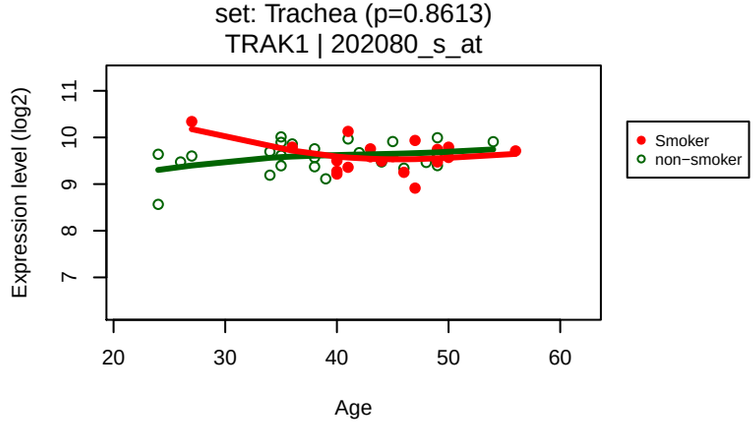

Gene expression ratio in Males
gene: TRAK1 | 202080_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0092
p=0.0695
p=0.389
p=0.1996
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRAK1 | 202080_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4702
p=0.0171
p=0.4869
p=0.5952
SmallAirway2
LargeAirway2
Trachea
Macrophage2

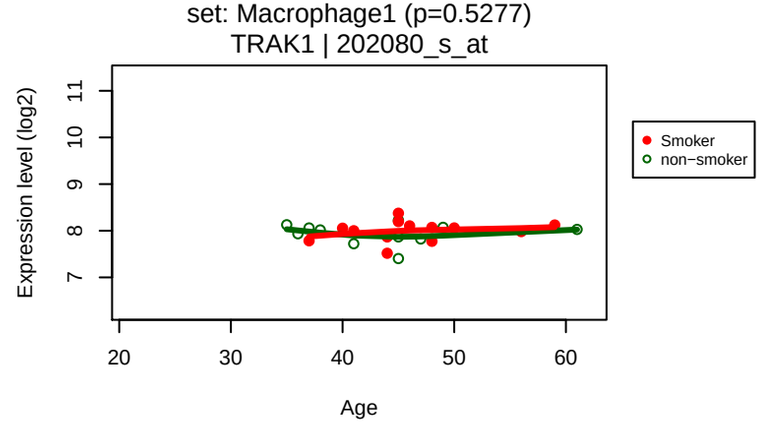

set: SmallAirway1 (p=0.0379)
TRAK1 | 202080_s_at
Expression level (log2)
Age
Smoker
non−smoker

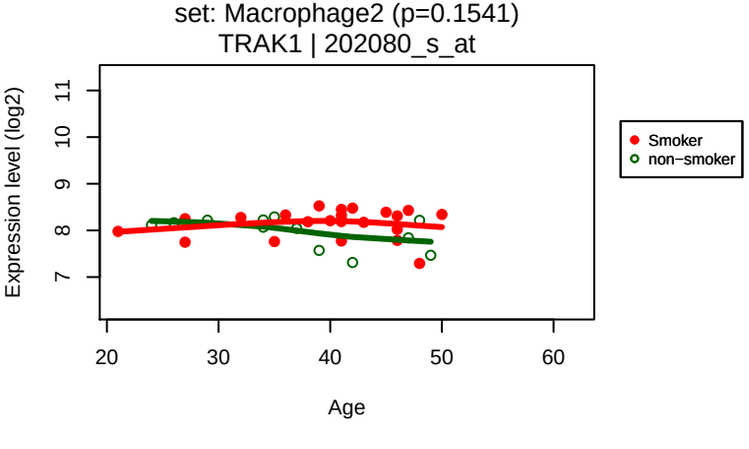

set: SmallAirway2 (p=0.0067)
TRAK1 | 202080_s_at
Expression level (log2)
Age
Smoker
non−smoker

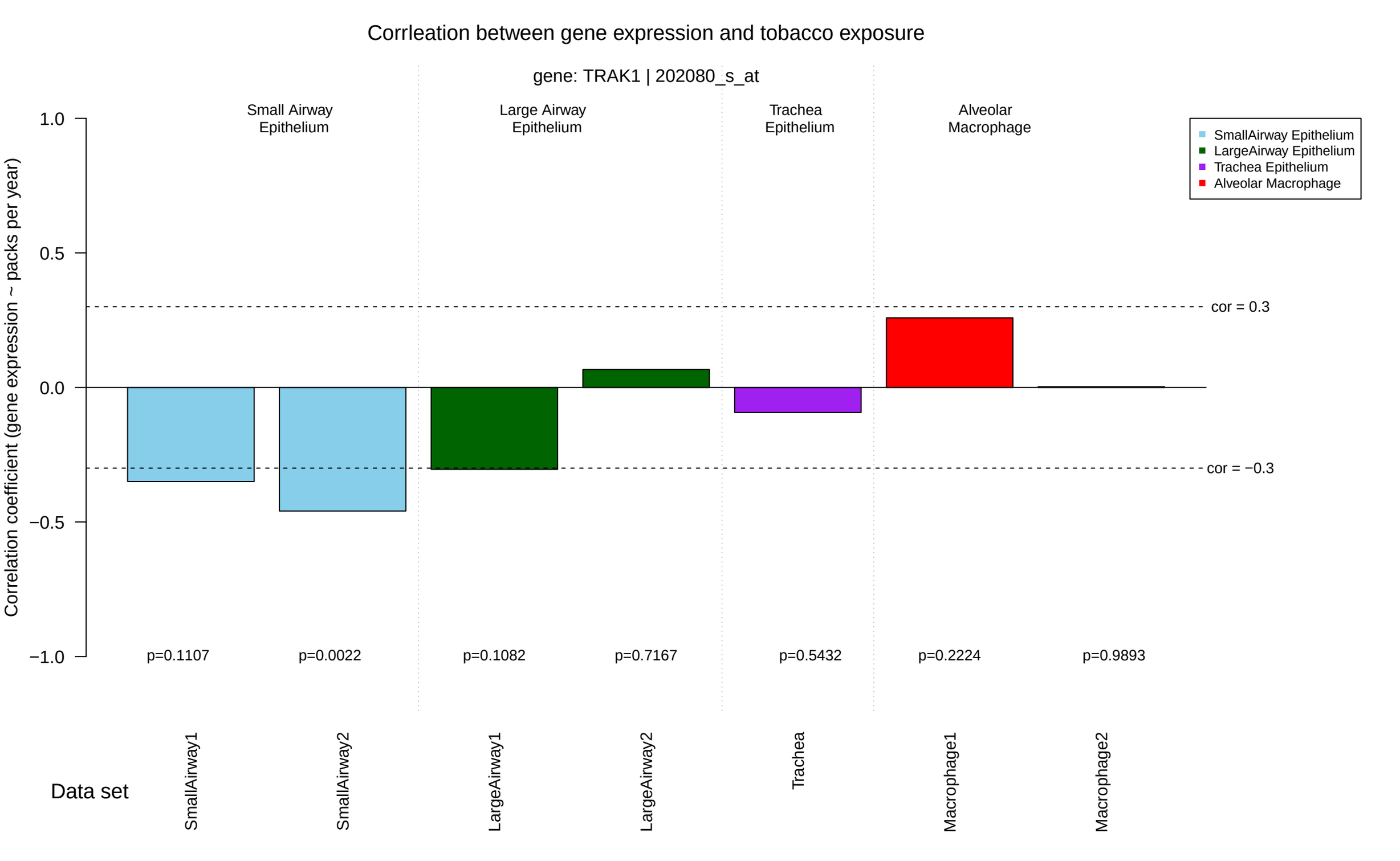

set: LargeAirway1 (p=0.0689)
TRAK1 | 202080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8863)
TRAK1 | 202080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.8613)
TRAK1 | 202080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5277)
TRAK1 | 202080_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.1541)
TRAK1 | 202080_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRAK1 | 202080_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1107
p=0.0022
p=0.1082
p=0.7167
p=0.5432
p=0.2224
p=0.9893
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2